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Manufacturing a Benchtop Urine Dipstick Analyzer (2P & 10P) for the Indian Market

A Comprehensive Technical, Manufacturing & Commercial Guide

Prepared for: Levram Life Sciences

Date: May 2026

Benchmarks: Siemens Clinitek Series | Mission Urine Analyzers

Classification: Confidential - Internal R&D & Business Development Document

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1. Executive Summary

Urinalysis is one of the most frequently ordered diagnostic tests globally, serving as a front-line screening tool for urinary tract infections (UTIs), diabetes, kidney disease, liver disorders, and metabolic conditions. In India, the urinalysis market is expanding rapidly, driven by rising diabetes and chronic kidney disease (CKD) prevalence, expansion of diagnostic laboratories, and the government's Ayushman Bharat program, which has accelerated diagnostics penetration into Tier 2, 3, and rural markets.

The current benchtop urine analyzer market in India is dominated by imported instruments from Siemens (Cinitex series), Mission Diagnostics, Roche (Cobas), and Dirui. These instruments are priced between ₹50,000-₹2,50,000 and rely on expensive proprietary reagent strips. There is a significant unmet need for a high-quality, economical, India-manufactured benchtop urine dipstick analyzer that can compete on accuracy, reliability, and after-sales service while offering strips at a lower cost.

This report provides a complete technical and commercial blueprint for Levram Life Sciences to develop, manufacture, and commercialize a 2-parameter (2P) and 10-parameter (10P) benchtop urine dipstick analyzer. The product leverages proven reflectance photometry technology, commercially available electronic components, and a modular design philosophy to achieve a target instrument manufacturing cost of ₹8,000-₹14,000 (2P) and ₹18,000-₹28,000 (10P), enabling retail pricing of ₹25,000-₹35,000 (2P) and ₹55,000-₹80,000 (10P) — well below the Siemens and Mission benchmarks.

2. Technology Overview & Principles

2.1 The Measurement Principle: Reflectance Photometry

Benchtop urine dipstick analyzers operate on the principle of **diffuse reflectance photometry**. When a urine dipstick is dipped into a urine sample and removed, the reagent pads undergo colorimetric reactions proportional to the concentration of the analyte. The analyzer illuminates each reagent pad with light of a defined wavelength and measures the intensity of light reflected back from the pad surface. The degree of color change on the pad alters the reflectance, which is then converted to a semi-quantitative result.

The fundamental relationship is described by the **Kubelka-Munk (K-M) equation**:

$$F(R) = (1 - R)^2 / 2R = K/S$$

Where: - **R** = measured reflectance (ratio of reflected to incident light) - **K** = absorption coefficient of the colored product - **S** = scattering coefficient of the pad matrix - **F(R)** = K-M remission function, linearly related to analyte concentration

In practice, instruments measure a **reflectance ratio** (sample pad reflectance vs. reference pad or direct light channel) and apply a calibration curve to map the ratio to a semiquantitative category (e.g., Negative / Trace / 1+ / 2+ / 3+) or a numerical concentration range.

2.2 Why Reflectance Photometry is Preferred

Feature	Reflectance Photometry	Transmission Photometry
Works with opaque matrices	Yes	No
Suitable for dry chemistry	Yes	No
Low-cost implementation	Yes	Moderate
Robust to strip-to-strip variation	Moderate	Low
Used in Siemens Clinitek	Yes	—

2.3 Optical Architecture Options

Three main optical architectures are used in commercial and research analyzers:

Architecture A - Direct LED-to-Photodetector (Single Point) - Simplest design; LED illuminates pad at fixed angle (typically 45°) - Photodetector at complementary angle collects reflected light - Used in low-cost portable readers - Sensitive to strip placement variation

Architecture B - Optical Fiber Bundle (Preferred for Benchtop) - LED light is conducted to the measurement zone via illumination fiber bundle - Reflected light is collected by a separate detection fiber bundle - Greatly reduces sensitivity to ambient light and strip placement - Used in commercial benchtop analyzers including Siemens Clinitek 50/100 - Recommended for Levram's design

Architecture C - Image Sensor / Camera Based - CMOS/CCD camera captures full-color image of all pads simultaneously - RGB or HSV analysis via embedded processor - Higher cost; used in high-throughput analyzers - Not recommended for first-generation economical product

2.4 Multispectral Illumination

Different reagent pads absorb at different wavelengths. A single white LED is insufficient for optimal signal-to-noise across all 10 parameters. The solution is **trichromatic or multispectral illumination**:

LED Color	Wavelength	Target Analytes
Green	525-565 nm	Hemoglobin, Bilirubin, Urobilinogen
Red/Orange	620-660 nm	Glucose, Protein, Leukocytes
Blue	430-470 nm	Nitrite, pH
Infrared	850 nm	Specific Gravity (reference)

The analyzer cycles LEDs sequentially (or uses optical filters) and records the photodetector response for each wavelength per pad. This enables accurate discrimination across all analytes.

2.5 Signal Processing Chain

Urine Strip → LED Illumination → Pad Reflectance → Photodetector → Transimpedance Amplifier → ADC → MCU (Dark Subtraction + Reference Ratioing + Temperature Compensation) → Calibration Curve Lookup → Semiquantitative Result → Display / Print / LIS Export

3. Strip Chemistry - 2P and 10P Parameters

3.1 The 2-Parameter Strip (2P)

The 2P strip is designed for high-volume, rapid screening — primarily for glucose and protein, the two most commonly screened analytes in India for diabetes monitoring and kidney disease screening.

Parameter	Reagent Chemistry	Color Change	Detection Range
Glucose	Glucose oxidase / peroxidase / chromogen (e.g., tetramethylbenzidine, TMB)	Yellow → Green → Blue	Negative to ≥ 2000 mg/dL
Protein	Tetrabromophenol blue (TBPB) with citrate buffer (protein error of indicators)	Yellow → Blue-Green	Negative to ≥ 500 mg/dL

Strip Construction (2P): - Polystyrene backing strip: 5 mm wide × 70–75 mm long
 - Two reagent pads (each ~5 × 5 mm), separated by 3–4 mm - One reference/blank pad (white cellulose) for background correction - Pads are cellulose fiber matrices impregnated with reagents and dried

3.2 The 10-Parameter Strip (10P)

The 10P strip is the standard clinical tool for comprehensive urinalysis. It measures:

#	Parameter	Reagent Chemistry	Color Range	Clinical Significance
1	Glucose	Glucose oxidase + peroxidase + TMB	Yellow → Green → Blue	Diabetes, renal glycosuria
2	Protein	Tetrabromophenol blue (TBPB), citrate buffer	Yellow → Blue-green	Nephritis, nephrotic syndrome
3	pH	Methyl red + bromothymol blue	Orange → Yellow → Green → Blue	Acid-base disorders, UTI
4	Specific Gravity	Bromothymol blue + poly(methyl vinyl ether/maleic acid)	Blue → Yellow-green	Hydration, renal concentration
5	Ketones	Sodium nitroprusside + glycine (Legal's reaction)	Buff → Maroon	Diabetic ketoacidosis, starvation
6	Bilirubin	2,4-Dichloroaniline diazonium salt	Buff → Tan-Brown	Liver disease, biliary obstruction
7	Urobilinogen	p-Dimethylaminobenzaldehyde (Ehrlich's reagent)	Colorless → Pink-Red	Hemolytic anemia, liver disease
8	Blood/Hemoglobin	Diethylamethylbenzidine (TMB) + peroxidase	Yellow → Green → Blue	Hematuria, hemoglobinuria
9	Nitrite	Griess reaction (sulfanilamide + NED)	White → Pink	Bacterial UTI (gram-negative)
10	Leukocytes	Indoxyl ester + diazonium salt (esterase reaction)	Buff → Purple	UTI, pyuria

3.3 Strip Procurement Strategy for India

Option A - Source Finished Strips (Recommended for Phase 1): - Source OEM strips from Chinese manufacturers (e.g., Dirui, URIT, Analyticon) - Validate strip-

analyzer compatibility with your optical system - Cost: ₹3–₹8 per strip (10P), ₹1.5–₹3 per strip (2P) at bulk volumes - Rebrand under Levram label (check regulatory requirements for strip registration)

Option B - Manufacture Strips In-House (Phase 2/3): - Requires reagent chemistry expertise, impregnation equipment, and lamination machinery - Capital investment: ₹40–₹80 lakh for strip manufacturing line - Enables full cost control and margin maximization - Requires separate CDSCO registration for the strip as an IVD reagent

Key Strip Suppliers (India-compatible): - Dirui Industrial Co., Ltd. (China) - widely validated - URIT Medical Electronic Co. (China) - cost-effective - Analyticon Biotechnologies (Germany) - premium quality - Erba Mannheim (Czech Republic/India distribution) - Tulip Diagnostics (India) - domestic option

4. Optical & Electronic System Design

4.1 Optical Module Design

LED Selection

LED	Wavelength	Part Example	Cost (INR)
Green LED	525 nm	Osram LT CPDP	₹8–₹15
Red LED	660 nm	Vishay VLCS5830	₹8–₹15
Blue LED	450 nm	Cree C503B-BAS	₹10–₹18
IR LED	850 nm	Osram SFH 4545	₹8–₹12
White LED (reference)	Broadband	Nichia NSPW500GS	₹5–₹10

- LEDs are driven by constant-current drivers (e.g., Texas Instruments OPT101 or discrete BJT circuits)
- PWM control via MCU timer allows intensity adjustment and sequential switching
- LED lifetime: >100,000 hours — essentially maintenance-free

Photodetector Selection

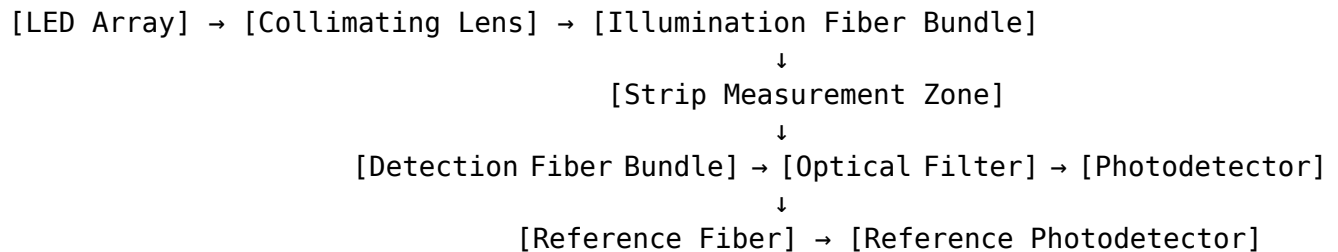
Component	Type	Part Example	Cost (INR)
Primary detector	Silicon photodiode	Hamamatsu S1087-01	₹80–₹150
Reference detector	Silicon photodiode	Hamamatsu S1087	₹80–₹150
Optional: array	TSL2591 light-to-digital	AMS TSL2591	₹60–₹100

Recommended configuration: - Two photodiodes: one for reflected light (measurement), one for direct/incident light (reference) - Ratiometric measurement: $R_{\text{measured}} / R_{\text{reference}}$ eliminates LED intensity drift - Optical bandpass filters (10 nm FWHM) placed in front of detectors for wavelength selectivity

Optical Fiber Bundle

- **Material:** Plastic optical fiber (POF), PMMA core, 0.5–1.0 mm diameter
- **Configuration:** Bifurcated bundle — one arm to LED, one arm to photodetector, common end at measurement zone
- **Numerical Aperture:** 0.5 NA for maximum light collection
- **Length:** 50–100 mm
- **Cost:** ₹200–₹500 per bundle (custom order, Chinese suppliers)
- Fiber bundles eliminate alignment sensitivity and ambient light interference

Optical Path Geometry



4.2 Analog Front-End Circuit

Transimpedance Amplifier (TIA) The photodiode generates a current proportional to incident light. A TIA converts this to a measurable voltage:

$$V_{\text{out}} = I_{\text{photo}} \times R_{\text{feedback}}$$

- **Op-Amp:** Texas Instruments OPA2134 (low noise, low offset) or OPA2340
- **Feedback resistor:** 1–10 MΩ (sets gain and bandwidth)
- **Feedback capacitor:** 1–10 pF (stability compensation)
- **Power supply:** ±5V or single 3.3V with rail-to-rail op-amp
- **Cost per TIA stage:** ₹25–₹60

Signal Conditioning

- Low-pass filter (RC, $f_c \sim 100$ Hz) to remove high-frequency noise
- Instrumentation amplifier for differential measurement where needed
- Voltage reference: **Texas Instruments REF3030** or **LM4132A-3.0** (3.0V, 0.05% accuracy) for ADC reference stability

Analog-to-Digital Conversion

Parameter	Specification
Resolution	16-bit (preferred) or 12-bit minimum
Sampling rate	1–10 ksps (more than sufficient)
Input range	0–3.3V or 0–5V
Channels	8–16 (for multi-LED, multi-detector)
Part example	ADS1115 (16-bit, I2C, ₹80–₹120) or STM32 internal 12-bit ADC

4.3 Microcontroller Unit (MCU)

Recommended MCU: STM32F103C8T6 (Blue Pill) or STM32F407VGT6

Feature	STM32F103 (2P model)	STM32F407 (10P model)
Core	ARM Cortex-M3, 72 MHz	ARM Cortex-M4, 168 MHz
Flash	64 KB	1 MB
RAM	20 KB	192 KB
ADC	12-bit, 10 channels	12-bit, 24 channels
UART/USB	Yes	Yes + USB OTG
Cost (INR)	₹120–₹200	₹350–₹600
SPI/I2C	Yes	Yes

MCU Firmware Functions: 1. LED sequential switching via GPIO/PWM timers 2. ADC sampling with averaging (16–64 samples per reading) 3. Dark current subtraction (reading with LEDs off) 4. Ratiometric calculation (sample/reference channels) 5. Temperature compensation using NTC thermistor data 6. Calibration curve lookup (stored in flash as polynomial coefficients or lookup tables) 7. Result classification to semiquantitative categories 8. UART/USB communication to display module or PC 9. Strip detection via IR sensor or mechanical switch 10. Self-test and error flagging routines

4.4 Display & User Interface

2P Model: - 2.4" TFT LCD (ILI9341 driver, SPI interface) — ₹250–₹400 - 4 tactile buttons (Start, Print, Menu, Power) - LED status indicators (Ready, Error, Processing)

10P Model: - 3.5" or 4.3" TFT LCD with touch (ILI9488 or RA8875 driver) — ₹400–₹700 - On-screen keyboard for patient ID entry - Optional: thermal printer integration (58mm paper, Bluetooth or serial) — ₹800–₹1,500

4.5 Power Supply

- Input: 100–240V AC, 50/60 Hz (India standard)
- SMPS module: 5V/2A + 12V/1A output — ₹200–₹400
- Onboard 3.3V LDO regulator (AMS1117-3.3) for MCU and analog circuits
- Optional Li-ion battery backup (18650, 2000 mAh) for portable use — ₹300–₹600
- Fuse + EMI filter on AC input for safety compliance

4.6 Communication Interfaces

Interface	Purpose	Implementation
USB Type-B	PC connectivity, data export	CH340G USB-UART, ₹20–₹40
RS-232	LIS/HIS connectivity	MAX232 level shifter, ₹15–₹25
Bluetooth 4.0	Wireless data to mobile app	HC-05 or HM-10 module, ₹80–₹150
SD Card	Local result storage	SPI SD card slot, ₹30–₹60

Interface	Purpose	Implementation
Optional: WiFi	Cloud connectivity	ESP8266/ESP32, ₹80-₹200

5. Mechanical Design & Housing

5.1 Housing Design Principles

The housing must: - Protect optical components from ambient light contamination - Allow easy strip insertion and removal - Be easy to clean (smooth surfaces, no crevices) - Look professional and clinical - Be manufacturable in India via injection molding or CNC

Recommended Material: ABS plastic (Acrylonitrile Butadiene Styrene) - Medical-grade, UV-resistant, easy to injection mold - Available in white/off-white for clinical aesthetic - Cost: ₹800-₹2,000 per housing (injection molded, 500+ units)

5.2 Key Mechanical Subsystems

Strip Insertion Channel

- A guided channel (slot) that positions the strip precisely under the optical head
- Tolerance: ± 0.2 mm lateral, ± 0.5 mm longitudinal
- Strip detection: IR break-beam sensor or mechanical microswitch triggers measurement sequence
- For 10P: motorized strip transport (small DC gearmotor, ₹150-₹300) for automated advancement

Optical Head Assembly

- Black anodized aluminum optical block houses fiber terminations, LED holders, and filter slots
- Machined to ± 0.05 mm tolerance for repeatable fiber-to-strip distance (typically 2-5 mm)
- Black felt/foam light seal around strip channel eliminates stray light

Thermal Management

- NTC thermistor (10 k Ω) mounted near optical block for temperature logging
- Passive ventilation slots; no active cooling needed for low-power LED system
- Operating range: 15-35°C (standard lab environment)

5.3 Dimensional Targets

Model	Dimensions (L $\times$ W $\times$ H)	Weight
2P Analyzer	180 $\times$ 120 $\times$ 80 mm	~600 g

Model	Dimensions (L × W × H)	Weight
10P Analyzer	280 × 180 × 120 mm	~1.2 kg

5.4 Industrial Design Reference

Benchmark the Siemens Clinitek Status+ (handheld) and Mission U120 for: - Strip channel design - Display placement - Button ergonomics - Cable routing

6. Bill of Materials (BOM) with Estimated Costs

6.1 BOM - 2-Parameter (2P) Analyzer

Category	Component	Qty	Unit Cost (INR)	Total (INR)
Optical	Green LED 525nm	2	₹12	₹24
	Red LED 660nm	2	₹12	₹24
	Silicon Photodiode (measurement)	1	₹120	₹120
	Silicon Photodiode (reference)	1	₹120	₹120
	Optical fiber bundle (bifurcated)	1	₹350	₹350
	Optical bandpass filters (2 wavelengths)	2	₹200	₹400
	Optical block (machined aluminum)	1	₹600	₹600
Electronics	STM32F103C8T6 MCU	1	₹160	₹160
	OPA2134 Op-Amp (TIA)	2	₹45	₹90
	ADS1115 16-bit ADC	1	₹100	₹100
	LM4132A voltage reference	1	₹35	₹35
	NTC Thermistor 10kΩ	1	₹10	₹10
	PCB (custom, 2-layer, 100×80mm)	1	₹250	₹250
	Passive components (R, C, L)	Lot	—	₹150
	Connectors, headers	Lot	—	₹80
	USB-UART (CH340G)	1	₹30	₹30
	2.4" TFT LCD (ILI9341)	1	₹320	₹320
Display/UI	Tactile buttons × 4	4	₹5	₹20
	LED indicators × 3	3	₹3	₹9
Power	SMPS 5V/2A module	1	₹280	₹280
	AMS1117-3.3 LDO	2	₹8	₹16
	Power connector, fuse, switch	Lot	—	₹60
Mechanical	ABS housing (injection molded)	1	₹900	₹900
	Strip channel guide (machined)	1	₹200	₹200
	IR strip sensor	1	₹25	₹25
	Screws, standoffs, labels	Lot	—	₹80
Miscellaneous	Thermal pad, foam seals	Lot	—	₹50
	Packaging (box, foam, manual)	1	₹180	₹180
TOTAL BOM (2P)				≈ ₹4,283
Assembly labor (2 hours @ ₹200/hr)				₹400
Testing & QC				₹300

Category	Component	Qty	Unit Cost (INR)	Total (INR)
Total COGS (2P)				≈ ₹4,983

Note: At 1,000 unit production volumes, economies of scale reduce BOM by ~15-20%, bringing COGS to approximately ₹4,200-₹4,500 per unit.

6.2 BOM - 10-Parameter (10P) Analyzer

Category	Component	Qty	Unit Cost (INR)	Total (INR)
Optical	Multi-LED array (Green, Red, Blue, IR)	4	₹14	₹56
	Silicon Photodiodes (measurement + reference)	4	₹120	₹480
	Optical fiber bundles (10 channels)	10	₹400	₹4,000
	Optical bandpass filters (4 wavelengths)	4	₹220	₹880
	Precision optical block (machined)	1	₹1,800	₹1,800
Electronics	LED driver ICs (constant current)	2	₹60	₹120
	STM32F407VGT6 MCU	1	₹500	₹500
	OPA2134 Op-Amp (TIA ×4 stages)	4	₹45	₹180
	ADS1115 16-bit ADC ×2	2	₹100	₹200
	REF3030 voltage reference	1	₹50	₹50
	NTC Thermistor	2	₹10	₹20
	Custom PCB (4-layer, 150×120mm)	1	₹600	₹600
	Passive components (R, C, L)	Lot	—	₹350
	Connectors, FFC cables	Lot	—	₹200
	USB-UART + RS232 (CH340G + MAX232)	1	₹60	₹60
	Bluetooth module (HC-05)	1	₹120	₹120
	SD card module	1	₹50	₹50
	4.3" TFT LCD touch (ILI9488)	1	₹650	₹650
	Thermal printer module (58mm)	1	₹1,200	₹1,200
	Tactile buttons × 6	6	₹5	₹30
Power	SMPS 5V/2A + 12V/1A	1	₹380	₹380
	AMS1117 LDO regulators	3	₹8	₹24
Mechanical	Battery backup (Li-ion 18650 + charger IC)	1	₹550	₹550
	ABS housing (larger, injection molded)	1	₹1,800	₹1,800
	Strip transport motor + mechanism	1	₹450	₹450
	Strip channel + guides	1	₹350	₹350
	IR sensors × 2	2	₹25	₹50
Miscellaneous	Screws, standoffs, labels, cable ties	Lot	—	₹150
	Thermal pads, foam seals, light baffles	Lot	—	₹120
	Packaging (box, foam, accessories, manual)	1	₹350	₹350
	TOTAL BOM (10P)			≈ ₹15,000
	Assembly labor (4 hours @ ₹200/hr)			₹800
	Testing & QC (extended)			₹600
	Total COGS (10P)			≈ ₹16,400

Note: At 500+ unit volumes, COGS for 10P reduces to approximately ₹14,000-₹15,500 per unit.

7. Manufacturing Process - Step by Step

7.1 Phase 1: PCB Design & Fabrication

Step 1.1 - Schematic Design - Use KiCad (free) or Altium Designer for schematic capture - Design analog front-end, MCU connections, power rails, LED drivers - Include test points on all critical nodes

Step 1.2 - PCB Layout - Separate analog and digital ground planes with single-point connection - Keep TIA circuits away from digital switching noise - Place decoupling capacitors (100 nF) at every IC power pin - Use 4-layer board for 10P model (signal, ground, power, signal) - Export Gerber files

Step 1.3 - PCB Fabrication & Assembly - Send Gerbers to PCB manufacturer: JLCPCB, PCBWay (China) or Sunshine Circuits (India) - Cost: ₹150-₹600 per board at 100-piece MOQ - SMT assembly: use JLCPCB's SMT service or local EMS in Pune/Bangalore - Through-hole components: hand-soldered in-house

Step 1.4 - PCB Testing - In-circuit test (ICT) with bed-of-nails fixture for production - Functional test: verify power rails, MCU boot, ADC readings, LED switching

7.2 Phase 2: Optical Module Assembly

Step 2.1 - Optical Block Machining - CNC machine optical block from black anodized aluminum (6061-T6) - Drill fiber holes to ± 0.05 mm tolerance - Tap LED mounting holes (M2 threads) - Source from local machine shops (Pune, Coimbatore, Ludhiana) — ₹600-₹1,800 per block

Step 2.2 - Fiber Bundle Preparation - Cut fiber bundles to required length - Polish fiber ends with 1 μ m lapping film (critical for light transmission efficiency) - Epoxy fiber ends into SMA or custom ferrules - Test transmission with optical power meter

Step 2.3 - LED & Filter Installation - Press-fit or epoxy LEDs into optical block - Install bandpass filters in filter slots with optical cement - Connect LED leads to PCB via flexible wire or ZIF connector

Step 2.4 - Photodetector Mounting - Mount photodiodes on PCB with precise alignment to fiber detection ports - Apply optical coupling gel (refractive index matching) between fiber end and photodiode active area - Shield with black felt to prevent crosstalk

Step 2.5 - Optical Module Characterization - Measure dark current (LEDs off): should be < 1 nA - Measure signal-to-noise ratio with reference white tile - Verify linearity with neutral density filters (0.1-2.0 OD range) - Record baseline reflectance values for calibration

7.3 Phase 3: Mechanical Assembly

Step 3.1 - Housing Preparation - Inspect injection-molded housing for warpage, sink marks, flash - Apply EMI shielding paint inside housing if required for regulatory

compliance - Install rubber feet, cable grommets, and ventilation grilles

Step 3.2 - Strip Channel Assembly - Install strip channel guide with alignment pins - Mount IR strip-detection sensor with precise positioning - Test strip insertion/detection with reference strips

Step 3.3 - Main Assembly - Mount PCB to housing with M3 standoffs (4 points) - Mount optical module to housing with alignment pins and 4 screws - Route and secure all cables with cable ties - Install display module with foam tape + screws - Install SMPS with strain relief on AC cable - Install front panel buttons and status LEDs

Step 3.4 - Final Mechanical Inspection - Check all screw torques - Verify strip channel alignment with optical head - Inspect for light leaks (test in darkened room with LED on) - Verify all external ports are accessible and labeled

7.4 Phase 4: Firmware Development & Loading

Step 4.1 - Firmware Architecture

main.c

— init.c	(hardware initialization)
— led_control.c	(LED PWM/GPIO driver)
— adc_driver.c	(ADC sampling, averaging, DMA)
— optical.c	(dark subtraction, ratiometry)
— temp_comp.c	(temperature compensation)
— calibration.c	(curve storage, lookup)
— classify.c	(result classification to categories)
— display.c	(LCD driver, UI state machine)
— comm.c	(USB, Bluetooth, RS232)
— storage.c	(SD card, result logging)
— selftest.c	(startup diagnostics)

Step 4.2 - Calibration Data Embedding - Calibration coefficients (polynomial or lookup table) for each analyte/wavelength combination - Derived from validation experiments with known concentration standards - Stored in MCU flash as const arrays

Step 4.3 - Firmware Loading - Use ST-Link V2 programmer (₹300-₹600) via SWD interface - Production programming: gang programmer for batch flashing - Lock flash memory after programming to protect IP

7.5 Phase 5: Calibration & Validation

(See Section 8 for detailed calibration protocol)

7.6 Phase 6: Final QC & Packaging

- Run full self-test routine
- Test with 3 known-concentration urine control samples (Negative, Low, High)
- Verify all 10 parameters (10P model) or 2 parameters (2P model) against expected results
- Record serial number, lot number, calibration date in instrument memory

- Apply CE/BIS labels, serial number sticker
 - Pack with power adapter, USB cable, quick-start guide, user manual, 5 test strips
-

8. Calibration & Quality Control

8.1 Calibration Philosophy

Calibration maps the raw reflectance signal (R) to a semiquantitative result category. It must account for: 1. **Lot-to-lot strip variation** (different reagent batches absorb differently) 2. **Instrument-to-instrument optical variation** (LED intensity, fiber transmission) 3. **Temperature effects** (reaction rates change with temperature) 4. **Aging effects** (LED intensity drift over time)

8.2 Factory Calibration Protocol

Step 1 - White Reference Calibration - Measure reflectance of a NIST-traceable white ceramic tile (R = 100%) - Set this as the 100% reflectance baseline for all channels - Store as factory calibration constant in MCU flash

Step 2 - Dark Reference - Measure photodetector output with all LEDs off - Store as dark offset; subtract from all subsequent measurements

Step 3 - Multilevel Calibration with Aqueous Standards - Prepare 5-7 concentration levels for each analyte (using certified reference materials) - Measure reflectance at each level - Fit polynomial calibration curve: $R = f(\text{concentration})$ - Store polynomial coefficients in MCU flash

Step 4 - Strip Lot Calibration - Each new lot of dipstick strips comes with a calibration code (printed on bottle or barcode) - Operator scans/enters lot code; instrument loads appropriate calibration offset - This compensates for inter-lot reagent variation (critical for accuracy)

Step 5 - Temperature Compensation - Measure instrument response at 15°C, 25°C, 35°C - Derive linear temperature correction coefficients for each analyte - Apply in firmware: $\text{Corrected_R} = \text{Measured_R} \times (1 + \alpha \times (T - T_{\text{ref}}))$

8.3 Field Calibration (User-Level)

- **Daily QC:** Run one Negative control and one Positive control strip each morning
- **Lot calibration:** Enter new lot code when opening a new strip bottle
- **Annual calibration:** Factory service or calibration kit with certified standards

8.4 Quality Control Program

QC Level	Frequency	Method	Pass Criteria
Daily QC	Every day of use	Negative + Positive control strips	All parameters within ± 1 grade
Lot-to-Lot	Each new strip lot	Run 3 controls from new lot	CV <10% vs. previous lot
Linearity	Monthly	5-point dilution series (glucose, protein)	$R^2 > 0.99$
Precision	Weekly	Repeat 10 $\times$ on same control	CV <5%
Accuracy	Quarterly	Compare vs. reference lab (Cobas/Clinitek)	$\geq 90\%$ agreement
Instrument QC	Annual	Full factory recalibration	Per manufacturer specs

8.5 Analytical Performance Specifications (Target)

Parameter	Sensitivity	Specificity	Precision (CV)
Glucose	$\geq 95\%$	$\geq 90\%$	<8%
Protein	$\geq 90\%$	$\geq 85\%$	<10%
Blood	$\geq 95\%$	$\geq 88\%$	<8%
Nitrite	$\geq 90\%$	$\geq 95\%$	<10%
Leukocytes	$\geq 85\%$	$\geq 90\%$	<12%
pH	± 0.5 pH units	—	<5%
Specific Gravity	± 0.005 SG units	—	<5%

9. Infrastructure & Equipment Required

9.1 Manufacturing Facility

Minimum Space Requirements: - Total area: 500-800 sq ft (can be expanded as production scales) - Clean assembly area: 200 sq ft (dust-controlled, positive pressure preferred) - PCB storage and components area: 100 sq ft - Testing & QC lab: 150 sq ft - Packaging & dispatch: 100 sq ft

Environmental Controls: - Temperature: 20-25°C (air conditioning) - Humidity: 40-60% RH (dehumidifier if needed) - ESD protection: anti-static flooring, wrist straps, ESD mats

9.2 Equipment List

Equipment	Purpose	Estimated Cost (INR)
Soldering station (Hakko FX-888D or equivalent)	PCB assembly	₹8,000–₹15,000
Hot air rework station	SMD rework	₹5,000–₹12,000
Digital multimeter (Fluke 87V)	Testing	₹12,000–₹18,000
Oscilloscope (100 MHz, 2-channel)	Electronics debug	₹15,000–₹35,000
ST-Link V2 programmer	Firmware loading	₹400–₹600
Optical power meter	Fiber/LED characterization	₹8,000–₹20,000
CNC milling machine (small desktop)	Optical block prototyping	₹50,000–₹1,50,000
3D printer (FDM, Creality Ender 3)	Prototype housings	₹15,000–₹25,000
Spectrophotometer (UV-Vis)	Reagent/strip validation	₹40,000–₹80,000
Analytical balance (0.1 mg)	Reagent preparation	₹20,000–₹40,000
pH meter (calibrated)	Strip chemistry QC	₹5,000–₹12,000
Refrigerator (4°C)	Strip/reagent storage	₹12,000–₹20,000
Laptop + KiCad/Altium license	PCB design	₹60,000–₹1,20,000
LCR meter	Component QC	₹8,000–₹15,000
Lapping/polishing kit	Fiber end polishing	₹3,000–₹6,000
ESD workstations × 3	Assembly	₹15,000–₹25,000
Total Equipment Investment		₹2,76,400 - ₹5,93,600

9.3 Human Resources

Role	Qualification	Number	Monthly Salary (INR)
Electronics Engineer	B.E./B.Tech Electronics	1	₹35,000–₹55,000
Embedded Firmware Developer	B.E. + C/C++ experience	1	₹40,000–₹65,000
Mechanical/Optical Engineer	B.E. Mechanical/Optical	1	₹30,000–₹50,000
QC Technician	B.Sc. / Diploma	1	₹18,000–₹28,000
Assembly Technician	ITI / Diploma	2	₹12,000–₹18,000 each
Regulatory Affairs Executive	M.Sc./M.Pharm + RA experience	1	₹30,000–₹45,000

Role	Qualification	Number	Monthly Salary (INR)
Total Monthly Payroll		8 people	₹1,77,000- ₹2,79,000

9.4 Software & Development Tools

Tool	Purpose	Cost
KiCad (EDA)	PCB schematic + layout	Free
STM32CubeIDE	Firmware development	Free
SolidWorks / FreeCAD	Mechanical design	₹0-₹2,50,000/year
MATLAB / Python	Calibration algorithm development	Free (Python)
LabVIEW / Python	Test automation	Free (Python)
Git + GitHub	Version control	Free

10. Regulatory Pathway in India (CDSCO / MDR 2017)

10.1 Regulatory Classification

Under the **Medical Devices Rules (MDR) 2017** and its amendments, a urine dipstick analyzer is classified as an **In Vitro Diagnostic (IVD) device**.

Classification: - **Instrument (Analyzer):** Class B IVD device (moderate risk)
- **Reagent Strips:** Class B IVD device (moderate risk) - Both require CDSCO registration before manufacturing and sale

10.2 Regulatory Pathway for Indian Manufacturer

Step 1 - Obtain Manufacturing License - Apply to State Licensing Authority (SLA) under Form MD-3 (for Class B devices) - Requires GMP-compliant manufacturing facility inspection - Fee: ~₹3,000-₹5,000 (state-level) - Timeline: 3-6 months

Step 2 - CDSCO Registration of the Device - File application on SUGAM portal (cdsco.gov.in) - Required documents: - Device Master File (DMF): design, specifications, manufacturing process - Risk Management File (per ISO 14971) - Performance Evaluation Data (analytical validation) - Biocompatibility data (ISO 10993) - Labeling and IFU (Instructions for Use) - Quality Management System certificate (ISO 13485 preferred) - Form: MD-7 (for new device registration by Indian manufacturer) - Fee: ₹2,500 per device (Class B) - Timeline: 6-12 months (CDSCO review)

Step 3 - ISO 13485 Certification (Recommended) - Quality Management System for medical devices - Required for export and preferred by large institutional buyers - Certification body: BSI, TÜV SÜD, Bureau Veritas - Cost: ₹3-₹8 lakh (initial audit + certification) - Timeline: 6-12 months

Step 4 - BIS Certification (If Applicable) - Bureau of Indian Standards (BIS) mark may be required for certain electronic medical devices - Check current mandatory BIS list for IVD devices - Apply under IS/IEC 61010-1 (safety for lab equipment)

Step 5 - Labeling Requirements Per MDR 2017 Schedule V, labels must include:
 - Device name, model number, lot number, serial number - Manufacturer name and address - Manufacturing date and shelf life - “For In Vitro Diagnostic Use Only” - CDSCO registration number - Storage conditions - Symbols per ISO 15223-1

10.3 Regulatory Timeline Summary

Activity	Duration	Cost (approx.)
Manufacturing license (SLA)	3-6 months	₹5,000-₹15,000
CDSCO device registration	6-12 months	₹25,000-₹75,000 (incl. documentation)
ISO 13485 QMS certification	6-12 months	₹3-₹8 lakh
Performance validation studies	3-6 months	₹2-₹5 lakh
Total Regulatory Timeline	12-24 months	₹6-₹14 lakh

10.4 Post-Market Requirements

- Maintain vigilance and adverse event reporting
- Annual renewal of manufacturing license
- Periodic device re-registration (as per CDSCO timelines)
- Maintain complaint log and corrective action records

11. Indian Market Analysis & Competitive Landscape

11.1 Market Overview

The Indian urinalysis market is one of the fastest-growing segments in the IVD diagnostics space:

- **Market Size (2024):** ~₹450-₹600 crore (instruments + strips)
- **Growth Rate:** ~14-16% CAGR (driven by diabetes, CKD, UTI burden)
- **Key Drivers:**
 - India has 101 million diabetics (IDF 2021) — largest in the world
 - CKD affects ~17% of Indian adults
 - Ayushman Bharat PM-JAY expanding diagnostics access
 - Growing diagnostic lab chains (Dr. Lal Path Labs, SRL, Thyrocare, Metropolis)
 - Increasing hospital-based lab automation

- **Key End-Users:**

- Hospital laboratories (tertiary, secondary)
- Standalone diagnostic laboratories
- Clinics and nursing homes (Tier 2/3 cities)
- Government health centers (PHC, CHC under NHM)
- Corporate health screening programs

11.2 Competitive Landscape

Competitor	Model	Technology	Price (INR)	Strip Cost	Market Position
Siemens	Clinitek Status+, Clinitek Advantus	Reflectance photometry, 10P	₹1,20,000–₹2,50,000	₹25–₹45/strip	Premium, dominant in large hospitals
Mission Diagnostics	U120, U500	Reflectance photometry, 10P	₹45,000–₹1,20,000	₹12–₹22/strip	Mid-market, strong in India
Roche	Cobas u 411	Reflectance photometry, 10P	₹1,80,000–₹3,00,000	₹30–₹55/strip	Premium, large hospitals
Dirui	H-100, H-500	Reflectance photometry, 10P	₹30,000–₹80,000	₹8–₹15/strip	Budget segment, growing
URIT	URIT-500B	Reflectance photometry, 10P	₹25,000–₹60,000	₹6–₹12/strip	Budget segment
Erba Mannheim	UriScan Pro	Reflectance photometry, 10P	₹50,000–₹1,00,000	₹10–₹20/strip	Mid-market
Levram (Target)	2P/10P Benchtop	Reflectance photometry	₹25,000–₹80,000	₹5–₹10/strip	Value-for-money, India-made

11.3 Levram's Competitive Advantages

1. **“Make in India” Positioning:** Growing preference for domestic products in government tenders (GeM portal)
2. **Lower Strip Cost:** By sourcing OEM strips from China and rebranding, or eventually manufacturing in-house, Levram can undercut imported strip prices by 30–50%
3. **Service Network:** Existing distribution network enables faster after-sales service vs. imported brands
4. **Customization:** Can offer vernacular language UI, India-specific report formats, and integration with local HIS/LIS systems

5. **Price Point:** Target pricing 30-50% below Siemens/Mission for comparable performance

11.4 Target Segments (Priority Order)

1. **Small-to-medium diagnostic labs** (Tier 2/3 cities) — highest volume, price-sensitive
2. **Nursing homes and clinics** — need simple, reliable 2P or 10P device
3. **Government health centers** — GeM portal procurement, tender-driven
4. **Corporate health screening** — bulk purchase, annual contracts
5. **Large hospital labs** — longer sales cycle, need ISO 13485 and performance data

12. Go-to-Market Strategy

12.1 Distribution Strategy

Levram's existing distribution network is the primary competitive moat. The recommended approach:

Tier 1 - Direct Distribution (Existing Network) - Leverage current distributor relationships in your territory - Train distributors on product demonstration, installation, and basic troubleshooting - Provide demo units to top 5 distributors in each region - Offer 20-25% distributor margin on instruments, 30-35% on strips

Tier 2 - Reagent Rental Model (Recommended for Market Penetration) - Offer the instrument free or at subsidized cost (₹5,000-₹10,000 deposit) - Lock customer to minimum monthly strip purchase (e.g., 200 strips/month) - Revenue model: ₹7-₹12 per strip × 200 strips × 12 months = ₹16,800-₹28,800/year per customer - This model is used successfully by Siemens and Mission in India

Tier 3 - Government/Tender Sales - Register on GeM (Government e-Marketplace) portal - Target CMSS (Central Medical Services Society) tenders - State-level NHM procurement for PHC/CHC labs - Requires CDSCO registration and ISO 13485

12.2 Pricing Strategy

Model	Instrument MRP	Strip MRP (per strip)	Strip MRP (per 100 strips)
2P Analyzer	₹24,999	₹4.50	₹399
10P Analyzer	₹64,999	₹9.50	₹899

Comparison vs. Benchmark: - Siemens Clinitek Status+ 10P: ₹1,80,000 instrument + ₹35/strip - Mission U120 10P: ₹75,000 instrument + ₹18/strip - Levram 10P: ₹64,999 instrument + ₹9.50/strip → **47% cheaper strips than Mission**

12.3 Launch Plan

Phase	Timeline	Activities
Phase 1: Development	Month 1-12	Prototype development, firmware, optical validation
Phase 2: Regulatory	Month 6-18	CDSCO filing, ISO 13485, performance studies
Phase 3: Pilot Launch	Month 18-24	50 units to key accounts, gather clinical feedback
Phase 4: Commercial Launch	Month 24-30	Full distribution rollout, GeM registration, marketing
Phase 5: Scale-Up	Month 30+	In-house strip manufacturing, export markets

12.4 Marketing & Promotion

- **KOL Engagement:** Partner with pathologists and lab directors in Tier 2/3 cities for clinical validation and endorsement
- **CME Programs:** Sponsor continuing medical education events for lab technicians and pathologists
- **Trade Shows:** Exhibit at MEDLAB India, Lab Asia, and NABL-accredited lab conferences
- **Digital Marketing:** LinkedIn, WhatsApp groups for lab professionals, YouTube demo videos
- **NABL Labs:** Target NABL-accredited labs first — they need documented, validated equipment

13. Cost of Goods & Pricing Strategy

13.1 Financial Summary

Metric	2P Model	10P Model
COGS per unit (100 units)	₹5,500	₹18,500
COGS per unit (500 units)	₹4,800	₹16,000
COGS per unit (1,000 units)	₹4,200	₹14,500
Target MRP	₹24,999	₹64,999
Distributor price (40% discount)	₹14,999	₹38,999
Gross margin at distributor price	~68%	~59%
Strip COGS (10P, 1000-strip MOQ)	₹3.50	—
Strip MRP	₹9.50	—
Strip gross margin	~63%	—

13.2 Break-Even Analysis

Fixed Costs (Annual): - Staff salaries: ₹25-₹35 lakh - Facility rent + utilities: ₹5-₹10 lakh - Regulatory + certification: ₹8-₹14 lakh (amortized over 3 years: ₹3-₹5 lakh/year) - Equipment depreciation: ₹3-₹5 lakh/year - **Total Fixed Costs: ~₹36-₹55 lakh/year**

Break-Even Volume (10P model at distributor price ₹38,999, COGS ₹16,000): - Contribution per unit = ₹38,999 - ₹16,000 = ₹22,999 - Break-even units = ₹45,00,000 / ₹22,999 $\approx$ **196 units/year**

This is a very achievable target — even 20 distributors selling 10 units each per year achieves break-even.

13.3 Recurring Revenue (Strips)

The real profitability in this business is in consumables (strips), not instruments:

- A lab running 20 tests/day $\times$ 300 days = 6,000 strips/year
- At ₹9.50 MRP and ₹3.50 COGS: **gross profit of ₹36,000/year per installed instrument**
- With 500 installed instruments: **₹1.8 crore/year in strip revenue**
- This is the “razor-and-blade” model used by Siemens and Mission

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End of Report